

Design and Validation of a Soft Switch for a Virtually Variable Displacement Pump

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Switch-mode hydraulic control is a compact and theoretically efficient alternative to throttling valve control or variable displacement pump control. However, a significant source of energy loss in switch-mode circuits is due to throttling during valve transitions. Hydraulic soft switching was previously proposed as a method of reducing the throttling energy loss, by absorbing, in a small variable volume chamber, the flow that would normally be throttled across the transitioning high-speed valve. An active locking mechanism was previously proposed that overcomes the main challenge with soft switching, which is a lock mechanism that releases quickly and with precise timing. This prior work demonstrated a reduction in energy losses by 66% compared to a control circuit. In this paper, a numerical model is developed for a switch-mode virtually variable displacement pump (VVDP) circuit, utilizing the proposed soft switch. The model is then used as a means of designing a proof of concept prototype to validate the model. The prototype design includes methods for controlling the soft switch spring preload, travel distance, piston displacement required to unlock the soft switch, valve command duty cycle, switching cycle period, and load pressure. Testing demonstrated that the soft switch circuit performed as expected in a baseline condition. The operating region for this prototype was found to be quite narrow. However, the model does a good job of predicting the displacement of the soft switch. [DOI: 10.1115/1.4038536]

Introduction and Background

In hydraulic power systems, two common methods of controlling the pressure or flow rate to the load are metering valves and variable displacement pumps. The use of metering valves to bleed off excess flow is inefficient, especially at low displacements, while variable displacement pumps are an expensive and bulky solution [1]. An alternative control architecture, the focus of this paper, rapidly switches the outlet flow from a fixed displacement pump to the load or tank via a high-speed valve. This approach uses a pulse-width-modulated (PWM) valve signal and is analogous to switch mode control in power electronics [2].

A simple hydraulic switch mode circuit utilizes a fixed displacement pump and a high-speed on/off valve to create a virtually variable displacement pump (VVDP) [1,3–5]. When the on/off valve is switched on, the full flow of the pump is sent to the load, and when it is off, the flow is sent to the tank. In both valve positions, the system experiences minimal throttling losses as the valve is fully open. By varying the percentage of time the on/off valve is switched to load (on), which is called the duty cycle, the flow rate reaching the load can be controlled. One of the main advantages of VVDPs is that the component cost has the potential to be lower than that of a variable displacement pump once high-speed switching valves become common commercial products.

Cao et al. show that five main loss mechanisms are active in switch mode circuits, namely, internal leakage, viscous friction, throttling during valve transitions, inertial forces of the high-speed valve, and fluid compressibility [6]. Rannow and coworkers determined that generally 60% of system losses are due to throttling during valve transitions, 20% is due to fluid compressibility, 19% is due to throttling when the valve is fully open, and 1% is due to valve leakage [3,7]. Leveraging the concept from power electronics, Rannow and Li proposed the use of a hydraulic soft switch, which is a method whereby fluid that would normally be throttled

across a control valve during valve transitions, instead flows into a small chamber with a spring-loaded piston, to be released after the valve transition is completed [8]. Since the flow rate across the control valve during transitions is lower, the soft switching concept can theoretically reduce throttling and compressibility losses by 81%.

The soft switch concept proposed by Rannow and coworkers is equipped with an externally actuated locking mechanism at an intermediate location of the piston travel [7]. The main difficulty of the soft switch approach is the design of the soft switch lock-release mechanism, which has to operate quickly during the correct part of the PWM cycle. Van de Ven proposed a novel lock-release mechanism that controls the unlocking of the piston through the balance of pressure forces on the front and back of the piston, removing the need for precise external electromechanical locking/unlocking components [9]. The proposed soft switch is shown in Fig. 1, including the lock-release mechanism. The piston is forced to the left end of the cylinder via a compression spring on the right side of the piston. There is an annulus in the middle of the piston that is internally ported to the right side of the piston. As the piston displaces to the right, the piston annulus aligns with an annulus in the cylinder and is then ported to tank. The magnet

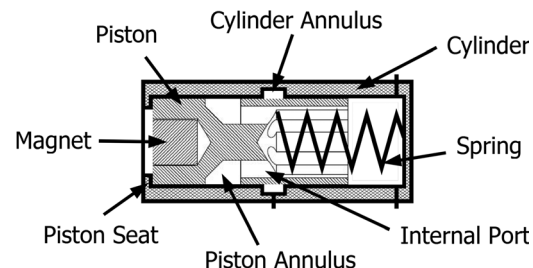


Fig. 1 Soft switch concept as proposed by Van de Ven (Adapted from Ref. [9])

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in the piston is used for experimental position measurement, as described in the physical design section.

The layout of the virtually variable displacement pumping circuit, which utilizes a fixed displacement pump and a high-speed three-way valve, is shown in Fig. 2. The left side or front side of the soft switch cylinder is connected to the switched volume of the system, the cylinder annulus is ported to tank, and the right or back side of the cylinder is ported to the load and tank lines via check valves.

The behavior of the soft switch during one cycle of high-speed valve PWM is as follows: The cycle begins with the high-speed valve sending fluid to tank and the soft switch empty with the piston seated to the left. When the three-way valve is transitioning from tank to load, the pressure in the switched volume increases as a result of throttling across the valve. This increased pressure in the switched volume moves the soft switch piston slightly to the right, compressing the spring and increasing the pressure of the fluid on the right of the piston. When the three-way valve finishes the transition to load, the piston remains stationary, except for slight movement due to leakage around the piston. As the three-way valve pressure port begins to close, the pressure in the switched volume becomes greater than that of the load line, which forces the pressure check valve to open and the soft switch piston to displace further to the right. When the piston has displaced enough for the annulus of the piston to align with the annulus of the cylinder, the piston is unlocked due to the decrease in pressure on the back side of the piston. The soft switch then rapidly absorbs the pump flow at tank pressure, which minimizes the throttling losses that the system would otherwise experience as the three-way valve transitions from load to tank. Once the three-way valve has switched completely back to tank, the spring on the back of the piston forces it back to its original position, sending the fluid inside the soft switch (on the left side of the piston) to tank. Simultaneously, fluid is drawn from tank into the back side of the soft switch through the tank check valve. With the piston reseated, the switching cycle can begin again.

Rannow and coworkers originally proposed a dual function soft switch that absorbs flow during both control valve transitions [7]. Van de Ven represented this idea in the proposition of two separate soft switches, an active and a passive soft switch [9]. The two soft switch chambers are placed in parallel in the switched volume of the circuit, with the passive switch designed to absorb pump flow during control valve transitions to load, and the active switch designed to unlock and absorb the pump flow during control valve transitions to tank. Operation of the active soft switch is the most challenging aspect of this approach, and thus, the purpose of this paper is to experimentally demonstrate the feasibility of the proposed unlocking concept. While prior simulations showed that using both the active and passive soft switches resulted in higher

system efficiencies, the passive soft switch will not be implemented in this work. To increase the chance of successful operation, the experimental active soft switch will be designed to be highly adjustable and at a larger scale than required for optimal efficiency to ease manufacturing.

The first section of this paper describes the development of a numerical model describing a PWM circuit operating with a soft switch. The Methods section outlines the design and testing of a physical system, including tuning parameters to explore the operating range of the soft switch. The numerical model and experimental results are then compared, followed by concluding remarks.

Numerical Model Development

Van de Ven outlined a numerical model of the soft switch with the proposed lock-release mechanism [9]. This section reviews and builds upon the previously presented model, showing the equations used to describe system port areas, pressures, flow rates, and component positions. Circuit component specifications, and when necessary dimensions, will be shown and their use in the model will be explained. The pump is assumed to be a constant flow source and the accumulator allows the load line to maintain constant pressure. All of the fluid conductors in the circuit are assumed to behave ideally, meaning that resistances and inertia are negligible.

The high-speed control valve used in the model is an off-the-shelf three-way valve (Hydroforce SV08-30) with circular tank ports and annular load ports. A graphical representation of spool displacement relative to tank and load port openings can be seen in Fig. 3, with the white moving plane representing spool displacement that allows the ports to open.

Since published valve information is limited, it is assumed that the valve spool displaces with a constant velocity, achieving the published valve transition times of 12 ms for the load port and 22 ms for the tank port. During spool transitions, there is a 127 μm valve overlap, where neither port is open. The orifice area of the load and tank ports can be described by

$$A_l = b_l \text{dis}_l \quad (1)$$

$$A_t = r_t^2 \cos^{-1} \left(1 - \frac{\text{dis}_t}{r_t} \right) + \left(\sqrt{(2r_t - \text{dis}_t)\text{dis}_t}(\text{dis}_t - r_t) \right) \quad (2)$$

where A_l and A_t are the effective orifice areas of the load and tank ports, respectively, b_l is the effective length of the base of the load port, r_t is the effective radius of the tank port, and dis_l , dis_t are the effective axial positions of the open area of the moving plane relative to the right edge of the load port and the left edge of the tank port, respectively. The tank and load port open areas for a switching period of 0.15 s, at a command signal duty cycle of 60%, are shown in Fig. 4. Note that the actual duty cycle of the control

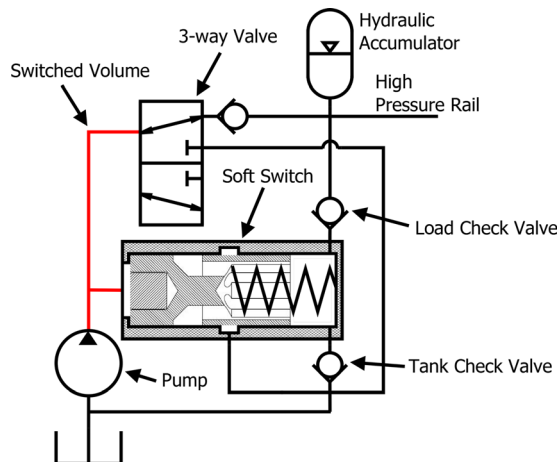


Fig. 2 Virtually variable displacement pumping circuit

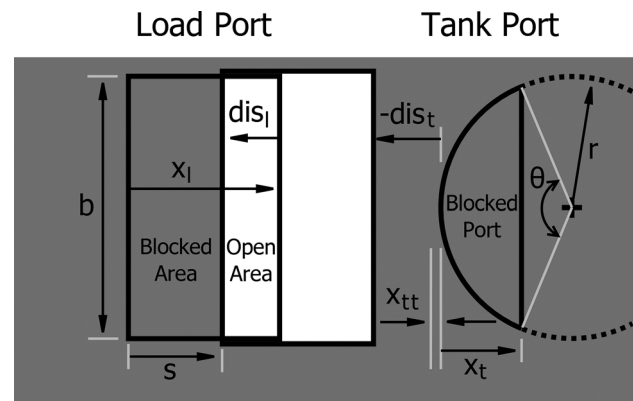


Fig. 3 Graphical representation of high-speed valve spool displacement relative to tank and load port openings

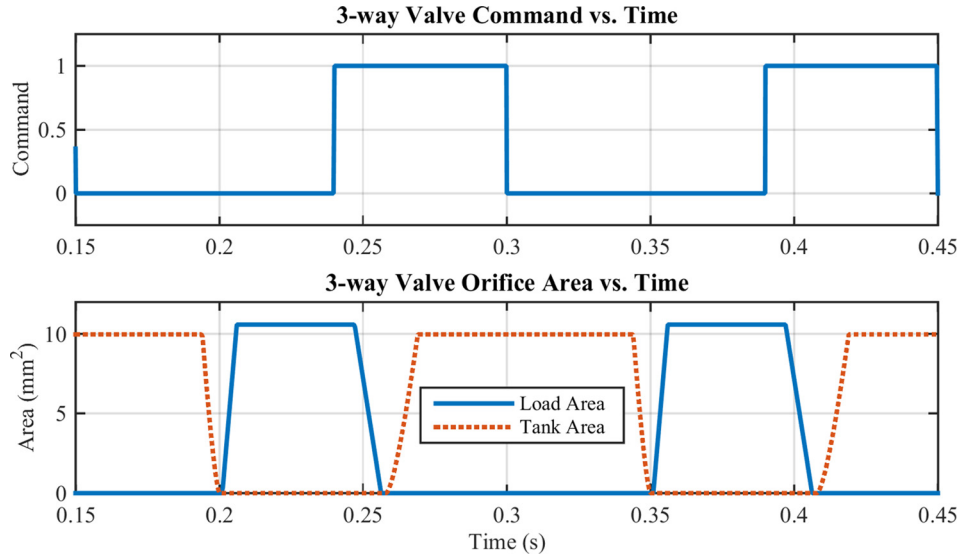


Fig. 4 Three-way valve orifice area as a function of time, with a 60% duty cycle and a period of 0.15 s

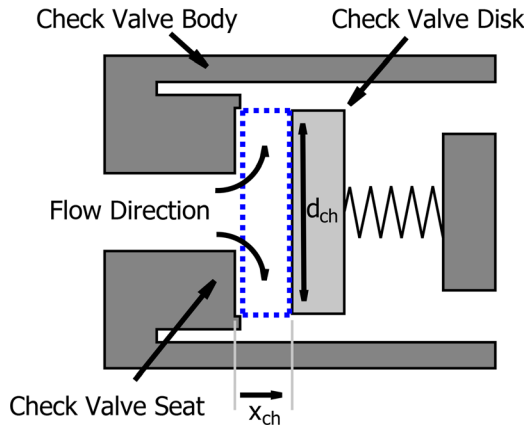


Fig. 5 Open area of check valves corresponds to the dotted cylindrical surface area between the check valve disk and seat

valve is less than the command signal duty cycle; due to the delay between the time the command signal is sent and the valve begins to transition to load being larger than the corresponding tank transition delay. These delays are based on the manufacturer's data-sheet for the valve used in the experimental studies.

Assuming that the open areas for the check valves are a cylindrical surface created by the displacement of the check valve disks, the open areas indicated by the dotted lines in Fig. 5, can be represented as

$$A_{ch,j} = \pi d_{ch,j} x_{ch,j} \quad (3)$$

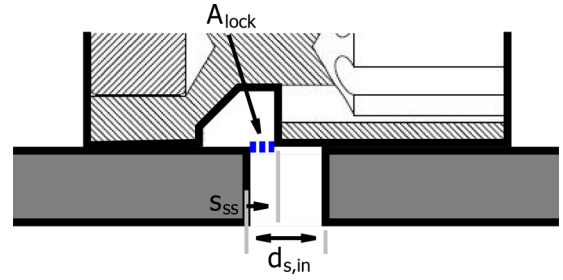


Fig. 6 Open area of the soft switch internal port to tank. The port is comprised of 6 radial holes through the cylinder sleeve.

where $d_{ch,j}$ is the effective diameter of the check valve disk, $x_{ch,j}$ is the disk displacement, and the subscript j is replaced by l and t for the load and tank, respectively. This approach introduces some error to the check valve area due to the small lip in the check valve body next to the disk seat, as shown in Fig. 5. Considering that the lip is only 2.5% of the maximum displacement of the check valve disk, the effects are negligible.

The soft switch piston port to tank, as shown in Fig. 6, is formed by an annulus on the piston moving past a set of radial holes equally spaced in the sleeve. When the soft switch piston port to tank begins to open, only a section of the circular ports is open, followed by the circular ports fully opening. Near the end of the piston displacement, the circular ports become partially blocked again. This behavior is described by the pseudocode

$$\begin{cases} \text{if } s_{ss} < 0 & A_{lock} = 0 \\ \text{else if } s_{ss} < d_{s,in} & A_{lock} = \left\{ \frac{3d_{s,in}^2}{4} \left[2 \cos^{-1} \left(1 - \frac{2s_{ss}}{d_{s,in}} \right) - \sin \left(2 \cos^{-1} \left(1 - \frac{2s_{ss}}{d_{s,in}} \right) \right) \right] \right. \\ \text{else} & A_{lock} = \left\{ \frac{3d_{s,in}^2}{4} \left[2\pi - 2 \cos^{-1} \left(3 - \frac{2s_{ss}}{d_{s,in}} \right) + \sin \left(2 \cos^{-1} \left(3 - \frac{2s_{ss}}{d_{s,in}} \right) \right) \right] \right. \end{cases} \quad (4)$$

where s_{ss} is the soft switch piston displacement referenced at the point where the edge of the piston annulus and cylinder port first align and begin to open to flow, A_{lock} is the area of the piston port to tank, and $d_{s,in}$ is the diameter of the cylinder port holes.

Since the hydraulic fluid is experiencing large pressure changes, it is appropriate to consider the compressibility of the fluid. The pressure dynamics for the switched volume and the volume behind the soft switch are derived from the definition of the bulk modulus, $\beta = -V(dP/dV)$. The effective bulk modulus, which takes into account the air entrained in the hydraulic fluid, is modeled using the relationship proposed by Cho et al. [10]

$$\beta_e(P) = \beta \left[\frac{\left(\frac{P}{P_0}\right)^{\frac{1}{\gamma}} e^{\frac{(P_0-P)}{\beta}} + R}{\frac{R}{\gamma} \frac{\beta}{P} + \left(\frac{P}{P_0}\right)^{\frac{1}{\gamma}} e^{\frac{(P_0-P)}{\beta}}} \right] \quad (5)$$

where $\beta_e(P)$ is the effective bulk modulus, P_0 is the atmospheric pressure, γ is the ratio of specific heats for air, and R is the volume fraction of air entrained in the oil. An expression for the time rate of pressure change in the switched volume can be found by rearranging the definition of flow rate, with the switched volume pressure

$$\dot{P}_{switch} = \frac{\beta_e(P_{switch})}{(A_p x_p + V_{switch})} (Q_{pump} - Q_{HSV,l} - Q_{HSV,t} - Q_{leak,f} - A_p v_p) \quad (6)$$

where the volume of interest is composed of the volume of fluid from the pump to the cylinder, V_{switch} , and the cylinder volume due to piston displacement; Q_{pump} is the flow rate out of the pump; $Q_{HSV,l}$ and $Q_{HSV,t}$ are the flow rates through the three-way valve to load and tank, respectively; $Q_{leak,f}$ is the leakage flow rate

from the switched volume to the volume behind the soft switch; and x_p is the displacement of the piston. Similarly, the rate of pressure change on the back side of the soft switch is described as

$$\dot{P}_{back} = \frac{\beta_e(P_{back})}{(V_{dead} + V_{back})} (Q_t + Q_{leak,f} - Q_l - Q_{lock} - Q_{leak,b} + A_p v_p) \quad (7)$$

here V_{dead} is the dead volume behind the piston, V_{back} is the changing volume behind the piston, Q_t and Q_l are the flow rates through the tank and load check valves, respectively, Q_{lock} is the flow rate through the locking port, and $Q_{leak,b}$ is the leakage flow rate from the back of the soft switch to tank.

The flow through the ports of the circuit is modeled using the orifice equation, which carries with it the limitations of the Bernoulli equation [11]. The Bernoulli equation assumes steady flow, negligible viscous effects and heat transfer, no shaft work, that the flow is incompressible, and that all of the flow follows the same streamline.

The flow rate through the high-speed valve tank port, using the variable area calculated in Eq. (2), is described as

$$Q_{HSV,t} = C_d A_t \sqrt{\frac{2}{\rho} |P_{switch} - P_{tank}| \text{sgn}(P_{switch} - P_{tank})} \quad (8)$$

where C_d is the discharge coefficient and P_{tank} is the pressure of the hydraulic reservoir.

As shown in Fig. 2, the circuit utilizes a check valve in series with the high-speed valve load port to ensure that flow reversals do not occur in the load line. For ease of simulation, the effects of the load port and the check valve on flow rate are combined, and the check valve is assumed to close instantaneously, as described by

$$\begin{cases} \text{if } P_{switch} > P_{load} + P_{ll} & Q_{HSV,l} = \sqrt{\frac{2(P_{switch} - P_{load} - P_{ll})}{\rho}} \frac{C_d A_l C_d A_{ch,ll}}{\sqrt{(C_d A_l)^2 + (C_d A_{ch,ll})^2}} \\ \text{else} & Q_{HSV,l} = 0 \end{cases} \quad (9)$$

where P_{load} is the constant load pressure, P_{ll} is the cracking pressure of the load line check valve, and $C_d A_{ch,ll}$ is the lumped discharge coefficient and area for the load line check valve based on the check valve specification sheet.

Two other check valves are used to connect the volume behind the soft switch piston to the tank and load lines. Like the high-speed valve ports, the flow through the check valves is modeled using the orifice equation and the check valve area, determined from the current position of the check valve (Eq. (3))

$$Q_l = C_d A_{ch,l}(x_{ch,l}) \sqrt{\frac{2|P_{back} - P_{load} - P_l|}{\rho}} \text{sgn}(P_{back} - P_{load} - P_l) \quad (10)$$

$$Q_t = C_d A_{ch,t}(x_{ch,t}) \sqrt{\frac{2|P_{tank} - P_{back} - P_t|}{\rho}} \text{sgn}(P_{tank} - P_{back} - P_t) \quad (11)$$

where P_l and P_t are the cracking pressures of the load and tank check valves, respectively.

The last port flow rate of interest in the soft switch circuit is through the soft switch locking port. One side of the port is connected to the back of the soft switch piston, while the other is connected to the tank, and the flow rate through it is described by

$$Q_{lock} = C_d A_{lock} \sqrt{\frac{2}{\rho} |P_{back} - P_{tank}| \text{sgn}(P_{back} - P_{tank})} \quad (12)$$

The piston uses a clearance seal, resulting in leakage between the piston and accompanying cylinder. The leakage flow paths past the piston are shown in Fig. 7. One of the paths, the dotted line on the left side of the piston, is leakage from the switched volume at the front of the piston to the volume behind the soft switch piston via the piston annulus, while the other leakage path, the dashed lines on the right side of the piston, is from the volume at the back of the soft switch to tank through two flow paths.

The leakage past the piston is modeled as incompressible laminar flow between parallel plates and is described as

$$Q_{leak,f} = \frac{2\pi d_p h_p^3 (P_{switch} - P_{back})}{3\mu x_{leak,f}} \quad (13)$$

where d_p is the diameter of the piston, h_p is half of the radial clearance between the piston and the cylinder, μ is the dynamic viscosity of the hydraulic fluid, and $x_{\text{leak},f}$ is the axial length of the leakage path at the front of the piston.

The leakage from the back of the piston to the internal tank port is quantified similarly, remembering that the number and the length of the leakage paths change with piston displacement

$$\begin{cases} \text{if } x_p < l_{\text{pre},r} + x_s & Q_{\text{leak},b} = \frac{2\pi d_p h_p^3 (P_{\text{back}} - P_{\text{tank}})}{3\mu x_{\text{leak},b,r} x_{\text{leak},b,l}} (x_{\text{leak},b,l} + x_{\text{leak},b,r}) \\ \text{else} & Q_{\text{leak},b} = 0 \end{cases} \quad (14)$$

where $l_{\text{pre},r}$ is the distance (relative to zero displacement of the sleeve) that the piston has to travel in order to open the port to tank, x_s is the displacement given to the sleeve relative to the left of the manifold, $x_{\text{leak},b,r}$ is the axial length of the leakage path at the back of the piston on the right side, and $x_{\text{leak},b,l}$ is the axial length of the leakage path at the back of the piston on the left side.

The soft switch piston undergoes motion due to the pressures on either side of the piston, the preload force of the spring behind the piston, the compression of the spring due to piston displacement, and the viscous friction related to the movement of the piston. The viscous friction term assumes laminar Couette flow between flat plates [11]. From a force balance, the acceleration of the piston is described by

$$\ddot{x}_p = \frac{(P_{\text{switch}} - P_{\text{back}})A_p - k_s x_p - F_s - \frac{\mu \dot{x}_p A_{p,\text{surf}}}{c_p}}{m_p} \quad (15)$$

where A_p is area of the piston face, k_s is the spring constant for the spring behind the piston, F_s is the force on the piston due to the spring preload, $A_{p,\text{surf}}$ is the surface area of the piston exposed to the cylinder surface, c_p is the clearance between the piston and the cylinder, and m_p is the mass of the piston.

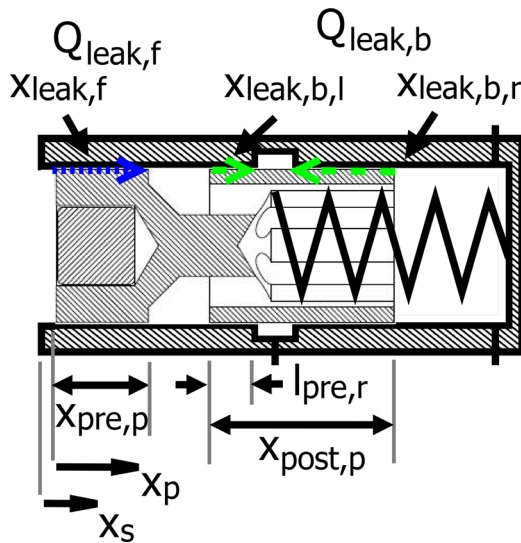


Fig. 7 Leakage flow paths past the soft switch piston. The path represented by the dotted line goes from the switched volume at the front of the piston to the volume behind the piston. The path represented by the dashed lines travel from the back of the soft switch to tank via two paths, as a result of the internal porting of the piston.

The physical travel limits of the soft switch piston are enforced by conditional statements on the displacement and velocity terms

$$\begin{cases} \text{if } x_p < 0 & x_p = 0 \\ \text{else if } x_p > x_{p,\text{max}} & x_p = x_{p,\text{max}} \end{cases} \quad (16a)$$

$$\begin{cases} \text{if } \dot{x}_p < 0 \text{ and } x_p \leq 0 & \dot{x}_p = 0 \\ \text{else if } \dot{x}_p > 0 \text{ and } x_p \geq x_{p,\text{max}} & \dot{x}_p = 0 \end{cases} \quad (16b)$$

where $x_{p,\text{max}}$ is the maximum displacement of the soft switch piston. This approach does not take into account the possibility of the piston bouncing upon impact with the seat and displacement limiter. The experimental results confirmed that this was a reasonable assumption, as described in the Results section.

The check valve disk accelerations are calculated using the same force balance approach used on the soft switch piston

$$\ddot{x}_t = \frac{(P_{\text{tank}} - P_{\text{back}})A_{d,t} - k_t x_t - P_t A_{d,t} - \frac{\mu \dot{x}_t A_{t,\text{surf}}}{c_t}}{m_t} \quad (17)$$

$$\ddot{x}_l = \frac{(P_{\text{back}} - P_{\text{load}})A_{d,l} - k_l x_l - P_l A_{d,l} - \frac{\mu \dot{x}_l A_{l,\text{surf}}}{c_l}}{m_l} \quad (18)$$

where A_d is the area of the disk face, k is the spring rate, x is the displacement of the disk, P is the cracking pressure, A_{surf} is the surface area of the disk exposed to the valve body, c is the clearance between the disk and the body, m is the mass of the check valve disk, and the subscripts l and t denote load and tank check valves, respectively.

The travel and velocity limits of the check valve disks are enforced as follows:

$$\begin{cases} \text{if } x_j < 0 & x_j = 0 \\ \text{else if } x_j > x_{j,\text{max}} & x_j = x_{j,\text{max}} \end{cases} \quad (19a)$$

$$\begin{cases} \text{if } \dot{x}_j < 0 \text{ and } x_j \leq 0 & \dot{x}_j = 0 \\ \text{else if } \dot{x}_j > 0 \text{ and } x_j \geq x_{j,\text{max}} & \dot{x}_j = 0 \end{cases} \quad (19b)$$

where l and t are substituted for j to represent load and tank, respectively, and $x_{t,\text{max}}$ and $x_{l,\text{max}}$ are the maximum displacements of the tank and load check valve disks, respectively. As with the modeling of the soft switch piston, this conditional statement approach does not take into account the possibility of the check valve disks bouncing upon impact with their physical limits. The oil was found to provide sufficient damping that bounce was not observed experimentally.

The parameters used to calculate port areas, pressures, flow rates, and component positions in the simulation of a VVDP

Table 1 The parameters used to calculate port areas, pressures, flow rates, and component positions for the soft switch simulations

Parameter	Symbol	Units	Value
System parameters			
Switching period	T	s	0.15
Duty cycle	duty	%	60
Dynamic viscosity of oil	μ	Pa·s	0.0275
Mass density of oil	ρ	kg/m ³	876
Tangent bulk modulus of air free oil	β	GPa	1.9
Ratio of specific heats for air	γ	unitless	1.4
Volume fraction of entrained air	R	%	4
Atmospheric pressure	P_0	kPa	101
Tank pressure	P_{tank}	kPa	827
Load pressure	P_{load}	MPa	5.62
Switched volume	V_{switch}	cm ³	30
Dead volume (behind piston)	V_{dead}	cm ³	1.87
Pump flow rate	Q_{pump}	l/min	3.786
Generic valve discharge coefficient	C_d	unitless	0.6
Three-way valve parameters			
Load transition time	tt_l	ms	12
Tank transition time	tt_t	ms	22
Effective base of rectangular load port	b_l	mm	16.0
Effective radius of circular tank port	r_t	mm	3.47
Spool port overlap	x_{tl}	μm	127
Discharge coefficient * area for load line C.V.	$C_d A_{\text{ch},ll}$	mm ²	3.3
Cracking pressure for load line C.V.	P_{ll}	kPa	124
Soft switch parameters			
Surface area of piston	$A_{p,\text{surf}}$	mm ²	790
Diameter of piston	d_p	mm	12.7
Piston mass	m_p	g	17.0
Radial piston/cylinder clearance	c_p	μm	31.2
Piston displacement to open tank port	$l_{\text{pre},r}$	mm	4.83
Maximum piston displacement	$x_{p,\text{max}}$	mm	12.7
Spring preload force	F_s	N	27.3
Spring rate	k_s	kN/m	5.38
Diameter of sleeve tank port holes	$d_{s,\text{in}}$	mm	5.59
Initial sleeve displacement	x_s	mm	2.54
Check valve parameters			
Actual area of load C.V. face	$A_{d,l}$	mm ²	26.1
Actual area of tank C.V. face	$A_{d,t}$	mm ²	48.7
Surface area of load C.V. disk	$A_{l,\text{surf}}$	mm ²	13.8
Surface area of tank C.V. disk	$A_{t,\text{surf}}$	mm ²	32.7
Effective diameter of load C.V. disk	$d_{\text{ch},l}$	mm	3.09
Effective diameter of tank C.V. disk	$d_{\text{ch},t}$	mm	5.47
Clearance of load C.V.	c_l	mm	1.44
Clearance of tank C.V.	c_t	mm	1.71
Mass of load C.V.	m_l	mg	155
Mass of tank C.V.	m_t	mg	502
Cracking pressure of load C.V.	P_l	kPa	103
Cracking pressure of tank C.V.	P_t	kPa	20.7
Spring rate of load C.V.	k_l	N/m	225
Spring rate of tank C.V.	k_t	N/m	170
Maximum displacement of load C.V.	$x_{l,\text{max}}$	mm	1.02
Maximum displacement of tank C.V.	$x_{t,\text{max}}$	mm	1.52

circuit with soft switch are summarized in Table 1. Using these parameters, the differential equations were solved using an Eulerian approach [12], which allowed for a simple implementation with reasonable accuracy for the small time-step used.

The operation of the baseline soft switch case is presented in Fig. 8. The second subplot shows the soft switch piston displacement. The piston displaces slightly when the switched volume pressure increases, period t_2 , and the fluid on the back side of the soft switch is compressed. Between periods t_2 and t_3 , the valve is connected to load pressure and the soft switch slowly displaces due to leakage. When the valve begins to close the pressure port, t_3 , the piston rapidly displaces and unlocks at a travel of 4.83 mm. When the piston unlocks, the switched volume pressure rapidly

drops, preventing throttling of flow across the transitioning valve. The soft switch piston returns to the base position at the end of the cycle, increasing the flow rate through the tank port.

Physical Design and Experimental Methods

This section presents the physical implementation of the soft switch concept. First, a description of the soft switch components and their interactions is shown. The experimental circuit setup is then described, followed by a description of the testing methods.

The soft switch design can be seen in Fig. 9 mounted directly to the three-way control valve manifold. The soft switch manifold has a main bore wherein the cylinder sleeve and piston are inserted. The piston is stainless steel and is free to travel axially within the cylinder between the stop on the left and the displacement limiting screw on the right. A compression spring provides a preload force to the piston that returns it to the seated position at the left of the cylinder. The piston is designed with internal porting to allow fluid to travel to the back side of the piston from the piston annulus, which, when aligned with the sleeve tank ports, performs the soft switch unlocking function. A magnet is embedded on the left side of the piston along the centerline to generate a magnetic field that is sensed by a magnetic displacement sensor to determine the piston position. The piston travels within the bronze sleeve, the position of which is adjusted by the placement of shims either in front of or behind the sleeve. This adjustment changes the piston displacement required to align the internal piston port to tank, and thereby, the soft switch unlocking distance. The load and tank check valves are placed near the rear of the aluminum manifold and ported to the back side of the piston through holes in the cylinder sleeve. Ports in both ends of the manifold for pressure transducers also serve the purpose of allowing air to be bled from the system.

The test stand for the soft switch circuit is shown in Fig. 10. It consisted of a 1.5 kW alternating current motor (A) driving a variable displacement axial piston pump (B) which had the pressure compensator defeated so that it behaves as a fixed displacement unit. The three-way valve and accompanying solenoid (C) had the normally open position switched to load. The soft switch manifold (D) was mounted directly to the three-way valve manifold, which was connected directly to the pump to reduce the system switched volume. The three-way valve load port connected to a check valve (E), a 1.0 L diaphragm accumulator (F) precharged to 4.27 MPa, a gear flow meter (R), a proportional pressure relief valve (G) powered by a 30V direct current (DC) power supply (H) that provided the system load, and a cooling coil (S), before returning to tank. Load and tank check valves (I) were located in the soft switch manifold and ported to the back of the soft switch to provide the unlocking signal and piston reseating flow, respectively. Located in the tank lines were the two ball valves (J and K) for bleeding air out of the system. A Coriolis flow meter (L) determined the flow rate into the pump, while a magnetic position sensor (M) determined the axial position of the soft switch piston. Three 20.7 MPa pressure transducers gathered switched volume pressure (N), pressure behind the soft switch piston (O), and load pressure (P), while a 20.7 MPa pressure gauge (Q) was used to visually determine cyclic steady-state operation. Component part numbers and specifications can be cross-referenced in Table 2, with clarification regarding the layout of the hydraulic portion of the test circuit provided by Fig. 11. The three-way valve is controlled via a transistor circuit (not pictured) emitting 24 V to the valve solenoid at the required duty cycle. The system temperature for all of the tests was maintained at $52.5 \pm 2.5^\circ\text{C}$. The fluid used in the test circuit was Mobil DTE 25 (ISO 46) which has a dynamic viscosity of approximately 0.0275 Pa·s at the operating temperature [13].

To determine whether the soft switch locking mechanism was functioning according to design, characteristics were developed that define proper operation:

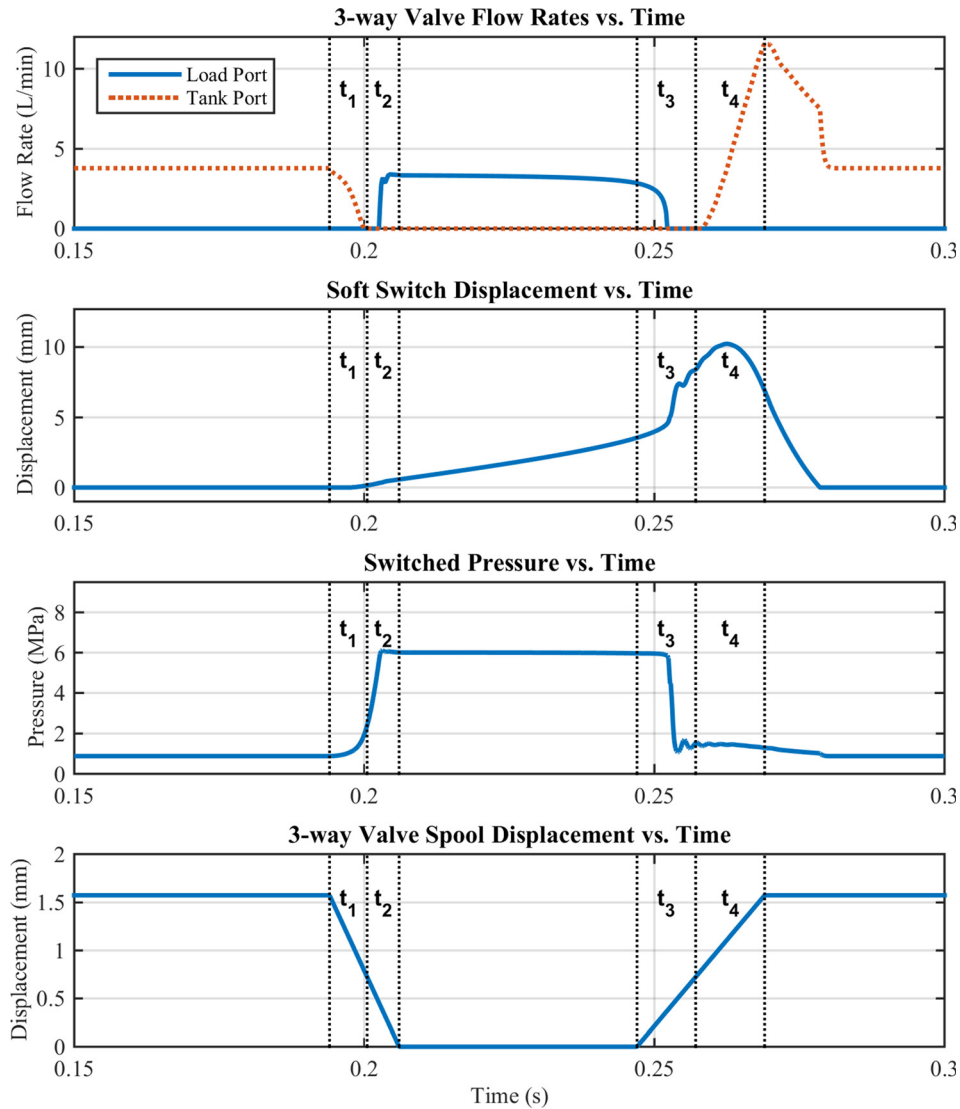


Fig. 8 Numerical model performance plots, showing the behavior of the soft switch for one complete switching cycle

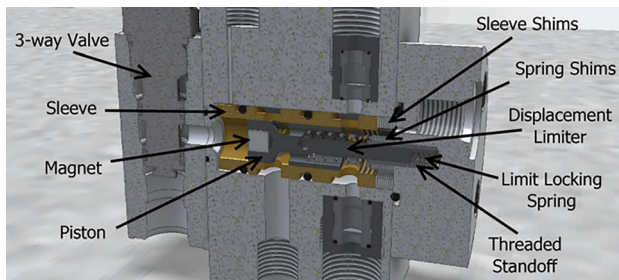


Fig. 9 Cross-sectional view of the assembled soft switch, including three-way control valve and manifold

- (1) The piston does not unlock too early or too late to absorb the pressure spike.
- (2) The piston does not reach the maximum displacement limit.
- (3) The piston reseats prior to the start of the next switching cycle.

The experimental soft switch was designed such that the valve duty cycle, switching period, soft switch unlocking distance,

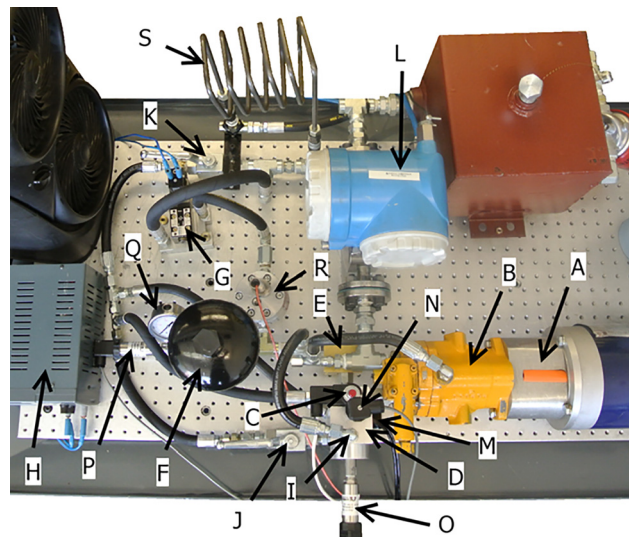


Fig. 10 Soft switch test bed setup with component labels

Table 2 Component part numbers and specifications for soft switch test bed components

Label	Name	Make	Model	Specifications
A	Motor	Industrial Duty	T2-18-56CB	2 HP, 1725 rpm
B	Variable displacement pump			11+ L/min
C	Three-way valve	Hydroforce	SV08-30	20.7 MPa
D	Soft switch			
E	Load line check valve	Eaton	CV6-10	76 L/min, 35 MPa
F	Accumulator	Parker	AD100B20T9A1	1 L, 20 MPa
G	Proportional pressure relief	Festo Didactic	151456	8 L/min, 12 MPa
H	DC power supply	Extech	382213	30 V/3 A
I _a	Load check valve	Inserta	ICS-C-02-N015	7.57 L/min, 103 kPa
I _b	Tank check valve	Inserta	ICS-C-04-N003	18.9 L/min, 20.7 kPa
J/K	Ball valve			
L	Coriolis flow meter	Endress + Hauser	Promass 80E08	10 MPa
M	Position sensor	Honeywell	HMC1512	120 mV
N/O/P	Pressure transducer	Omega	PX4201	20.7 MPa, 100 mV
Q	Pressure gauge			20.7 MPa
R	Gear flow meter	AW Gear Meters	JVA-20KG-25	7.57 L/min, 34.5 MPa
S	Cooling coil			

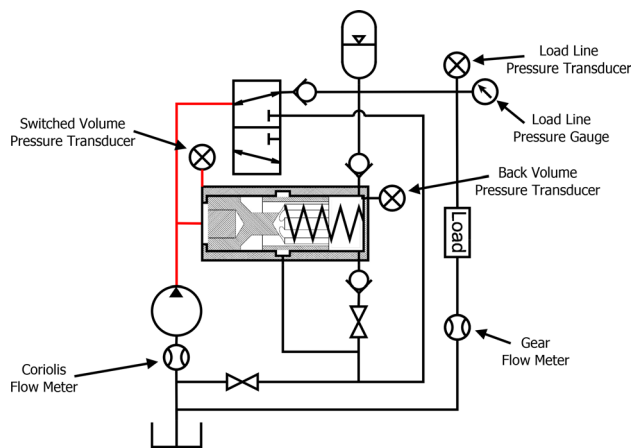


Fig. 11 The soft switch hydraulic test circuit layout. Three pressure transducers and two flow meters gathered pertinent information regarding system behavior.

maximum soft switch displacement, and soft switch spring pre-load were adjustable. To define a baseline case, Table 3, the physical soft switch parameters were placed near the middle of their adjustment ranges, and a pump flow rate and a load pressure were found where the soft switch behaved properly. For comparison, a control case was defined, also presented in Table 3, where the soft switch was held in the locked position and the valve duty cycle was adjusted to create the same pump flow rate as the baseline soft switch case. The valve duty cycle adjustment is necessary because a small portion of the flow that normally goes to load in the circuit without a soft switch will be absorbed by the soft switch and then flow to tank. In order to directly compare the circuits, a flow duty ratio is defined as the flow rate reaching the load divided by the flow rate of the pump. The flow duty ratio is maintained at the same value for all cases through the adjustment of the valve duty cycle.

To explore the behavior of the soft switch under varying operating conditions, experiments were run with the load pressure and valve switching period individually adjusted to a high and low condition. The high and low values were selected such that they

Table 3 Operating parameters for the experimental cases

Parameter	Symbol	Units	Value					
			Control case	Baseline case	High pressure	Low pressure	High period	Low period
Duty cycle	duty	%	52	60	63	52	58	68
Flow duty ratio	duty _f	%	30.3	30.3	30.3	29.8	30.2	30.7
Load pressure	P _{load}	MPa	5.63	5.62	5.72	5.52	5.62	
Valve period	T	s	0.15	0.15	0.15		0.152	0.138

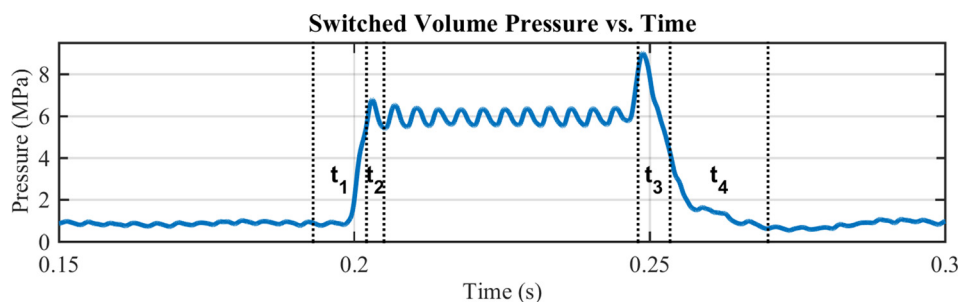


Fig. 12 Experimentally measured switched volume pressure profile for control case with the soft switch locked. The switched volume pressure spike reaches 9.02 MPa.

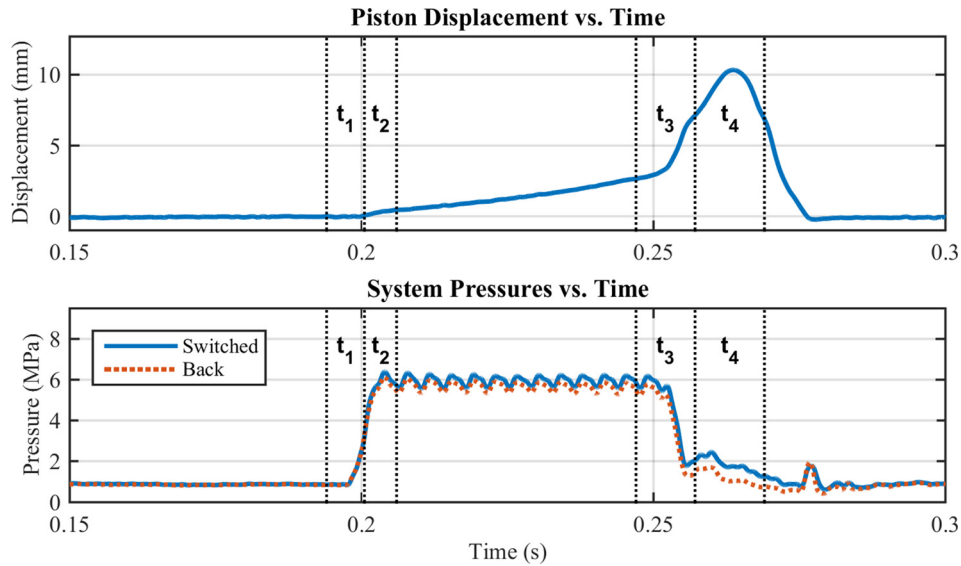


Fig. 13 Experimentally measured soft switch piston displacement and pressure profiles for baseline parameters

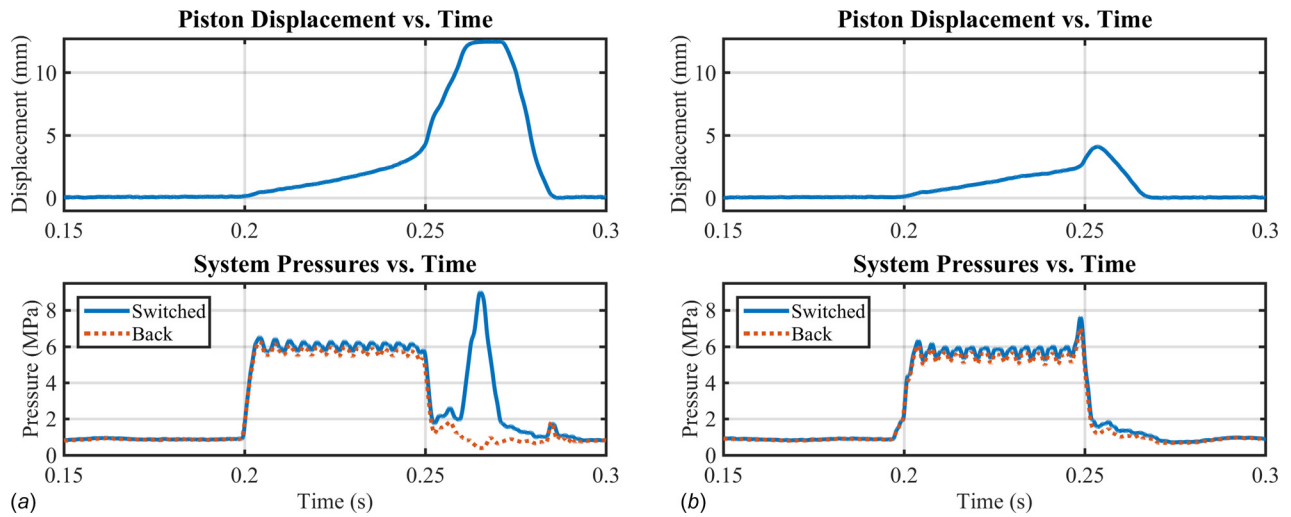


Fig. 14 Experimentally measured high and low system load pressure cases resulting in operation failure: (a) high load pressure of 5.72 MPa and (b) low load pressure of 5.52 MPa

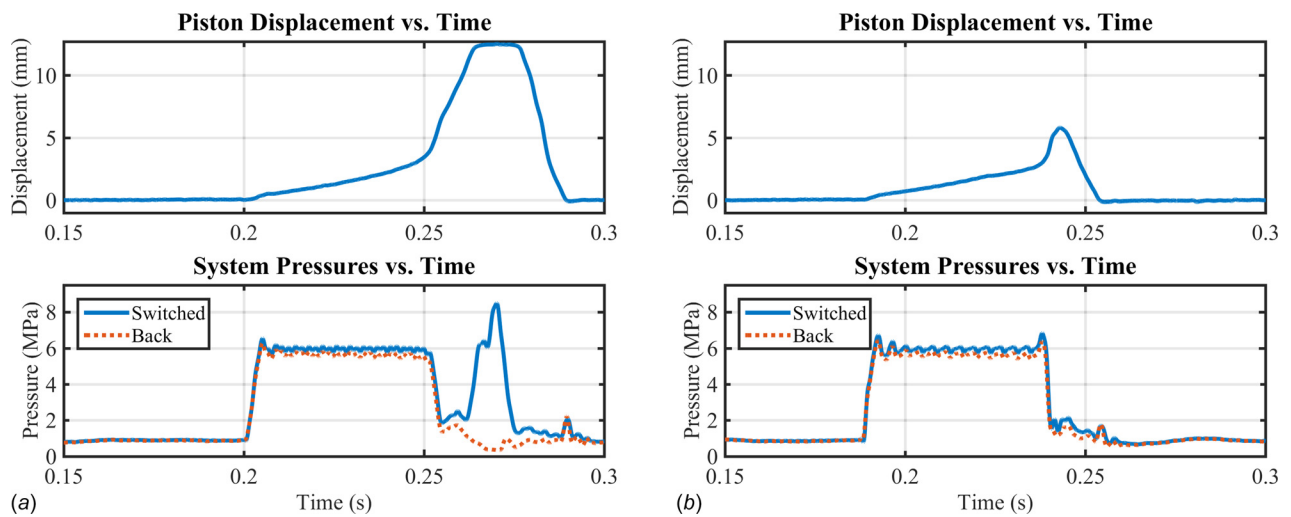


Fig. 15 Experimentally measured high and low valve switching period cases resulting in operation failure: (a) high valve switching period of 0.152 s and (b) low valve switching period of 0.138 s

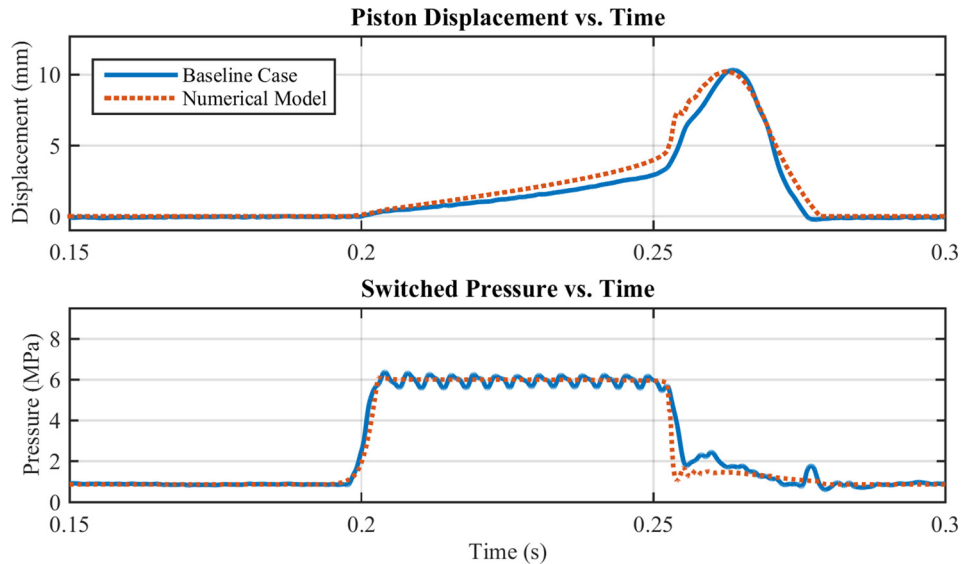


Fig. 16 Comparison between the experimental measurements and the numerical model for the baseline case

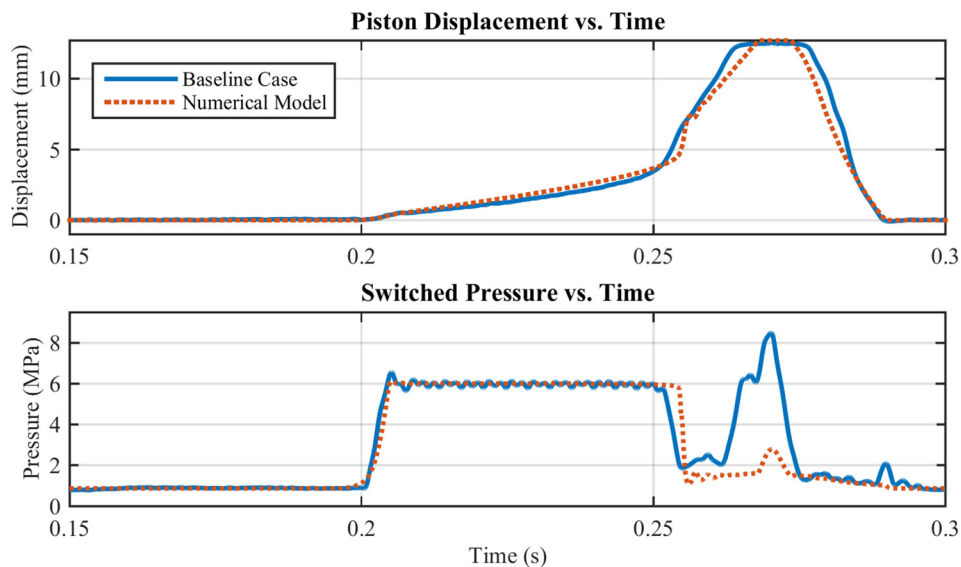


Fig. 17 Comparison between the experimental measurements and the numerical model for the high switching period case. The switched volume pressure is not accurately predicted due to slow unlocked piston movement.

just began to cause the soft switch operation to fail due to one of the conditions previously described, hence defining the operation bounds. For each operating case, the valve duty cycle was adjusted such that the flow duty ratio matched the control and baseline cases. A summary of the operating conditions can be seen in Table 3.

Results

This section will present experimental and simulation results for the soft switch circuit. Values for the adjustable circuit parameters that allow for proper experimental operation will be outlined and comparisons will be made to the numerical model.

A comparison for the soft switch cases is provided by the control case, where the soft switch piston was locked at a displacement less than the cracking distance. The switched volume

pressure profile for the control case is shown in Fig. 12. Four time intervals are shown in the figure, relating to the estimated position of the control valve during the switching cycle. Time period t_1 represents the control valve tank port closure, t_2 represents the opening of the control valve load port, t_3 indicates the control valve load port closure, and t_4 represents the opening of the control valve tank port. The pressure in the switched volume builds as expected when the control valve switches to load, and the pressure spikes to 9.02 MPa during the transition back to tank. Note that at higher pump flow rates, a pressure spike would also occur during the transition to the load branch. The oscillation in the switched volume pressure between time periods two and three is due to the flow ripple of the axial piston pump.

The piston displacement and pressure profiles of the baseline soft switch case are shown in Fig. 13, where the labels of switched and back refer to the measured pressure in the switched volume and the back of the soft switch piston. Notice that when the

switched volume pressure begins to build, as a result of the control valve transition to load, the soft switch piston experiences an initial displacement, followed by a fairly steady displacement due to leakage, until the control valve switches back to tank. During the control valve transition to tank, the piston displaces to the unlock position, and no pressure spike occurs. At the end of the control valve transition, the switched volume returns to tank pressure, allowing the soft switch piston to reseat in preparation for the next cycle.

The cases where the load pressure and valve switching period were set to high and low values were designed such that the soft switch fails to operate correctly by either unlocking early or not unlocking. The behavior of the high and low load pressure cases is presented in Fig. 14, while the high and low valve switching period cases are shown in Fig. 15. In both the high pressure and high period cases, the leakage past the soft switch piston causes the piston to reach the cracking threshold of 4.83 mm when the control valve is still connected to load, creating a sudden drop in pressure as the soft switch is filling and then a pressure spike when the piston reaches the travel limit. In the low pressure case, the pressure rise during the transition to tank is not sufficient to overcome the spring force to create the required displacement to unlock the soft switch. In the low period case, while the soft switch does unlock, it occurs late in the cycle resulting in little absorption of flow during the valve transition.

A comparison between the model predicted and experimentally observed piston displacement and switched volume pressure for the baseline is shown in Fig. 16. Note that the pump is assumed to be a constant flow source in the model, and thus, the pressure ripple generated by the physical pump is not predicted. The same comparison for high switching period case where the piston unlocks early is presented in Fig. 17.

Discussion and Conclusion

The experimental studies met the objective of successfully demonstrating the active soft switch. Under the correct system parameters, the soft switch unlocked at the correct time in the cycle and absorbed flow during the valve transition to minimize throttling across the control valve and eliminate the pressure spike. While not quantitatively measured, soft switching also reduced the audible noise. The high and low conditions of the load pressure and valve switching period were used to define the bounds for successful operation of the soft switch, which were found to be quite narrow. This narrow band is due to two primary reasons. First, the pressure spike during the control valve transition to tank in the control (no soft switch) circuit is quite small compared to most switch-mode circuits. This is due to the large flow area of the available control valve for relatively low flow rate of the available pump and the large switched volume. This required a low rate spring behind the soft switch piston to unlock with a small pressure differential, essentially creating a light trigger with little room for error. Second, the soft switch dimensions were oversized for manufacturing ease and are nonoptimal, which made the soft switch more sluggish to respond to pressure differentials.

The challenge of the light trigger behavior was compounded by the large manufacturing tolerances that resulted in a large soft switch piston-cylinder gap, causing significant leakage past the piston. This leakage allowed the piston to displace when the circuit was connected to load. As seen in the baseline case, Fig. 13, the lock release is not due to an increase in the switched volume pressure, but simply an alignment of the timing of the cycle with the soft switch piston displacement due to leakage. A lock-release as designed is better represented by the low switching period case,

Fig. 15(b), where a pressure rise triggers release of the soft switch.

As evident in Fig. 16, the model does a good job of predicting the soft switch displacement and the pressure in the switched volume for the control case. A small discrepancy in the model is a prediction that the piston displacement occurs slightly faster than observed experimentally, which led to a slightly early prediction of a drop in pressure indicating the soft switch unlock. For the high switching period case, Fig. 17, the model also does a good job predicting the soft switch piston initial displacement and unlock timing. However, the model predicts the soft switch to be near the maximum displacement for a shorter amount of time than observed experimentally. This leads to an inaccurate prediction of the switched volume pressure spike magnitude and timing. This model agreement is representative of the other cases as well, where good agreement is found for the piston displacement, but small deviations in maximum travel timing, resulted in errors in the pressure prediction.

In future experimental validation of the unlocking soft switch concept, more care needs to be given to maintaining the gap between the piston and sleeve, perhaps through hardened surfaces, and piston centering through multiple lands and grooves around the piston. Reducing the piston-sleeve clearance would reduce the leakage and extend the operating range. Knowledge of the position of the control valve spool would also assist in improving the capabilities of the numerical model, and understanding the specific timing observed experimentally. One option for quantifying the piston position is to measure the current flowing through the control valve solenoid and utilize the solenoid valve displacement predication scheme outlined by Yudell and Van de Ven [14]. Now that the concept has been validated and the foundation for a design model has been constructed, the soft switch design can be optimized to expand the operating range.

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